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Joseph Henry Briggs

Day Month Year

Contents

1	Characterisation of InGaAs Photodiodes for detecting 2μm Light	5
1.1	Good references	5
1.2	Introduction	5
1.3	What is an Extended InGaAs Photodiode	7
1.4	What is Needed from the Read-out Photodiode in a Gravitational Wave detector	10
1.5	The effect of reverse bias on an Extended InGaAs Photodiode -FD10D	12
1.5.1	Apparatus	12
1.5.2	Method	13
1.5.3	Results	15
1.5.4	Saturation Limit for some laser components photodiodes . . .	18
1.6	Dark Current and Dark Noise in Extended InGaAs Photodiodes . . .	19
1.6.1	Noise in photodiodes	20
1.6.2	Room Temperature	21
1.6.3	Cooling Experiment	21
1.6.4	Forward Bias Characteristics of IG24	23
1.7	InSb testing	25

1.8	Discussion	25
1.9	Conclusion	25
A	Extended InGaAs Diodes Noise Spectrums	26
A.1	Bias Section	26
A.2	Temperature Section	31
B	Photodiode Circuits	35
B.1	Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes	35
B.1.1	Transfer function	38
B.1.2	Noise measurement	40
B.2	Transimpedance Amplifier for Making Shot Noise limited Measure- ments of 10 mA Photocurrents	40
B.3	Simulations with LISO	43
B.4	Transfer Function Calculation With MATLAB	43

Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter 1: Significant amounts of thought has already gone into this.
- Chapter ??: Not started. Will write last/almost last

Chapter 1

Characterisation of InGaAs

Photodiodes for detecting $2\mu\text{m}$ Light

1.1 Good references

- 95% QE diode with cut-off at $2.4\mu\text{m}$ [1].
- semiconductor book [2]
- photodiode/optics book [3]

1.2 Introduction

The current generation of gravitational will be limited by the thermal noise due to brownian motion of the molecules that make up the mirror coatings. Since these thermally driven fluctuations are a limiting source of noise [4], proposed future gravitational wave detectors such as the Einstein Telescope [5] and LIGO Cosmic

Explorer [6] will use cryogenically cooled mirrors.

However, cryogenically cooled mirror coatings will not be compatible with the current laser wavelength, which is $1\text{ }\mu\text{m}$, and so future gravitational wave detectors may use either a $1.5\text{ }\mu\text{m}$ or $2\text{ }\mu\text{m}$ wavelength laser. Currently, $2\text{ }\mu\text{m}$ is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength.

Currently, as InGaAs is widely used semiconductor for detecting $1\text{--}1.6\text{ }\mu\text{m}$ light and thus suitable InGaAs photodiodes were available for detecting the gravitational-wave-signal-containing laser beam that emerges from the OMC. These photodiodes are low noise, linear, can detect many tens of milliwatts of light, and have a high quantum efficiency. Additionally, InGaAs is a popular material to use as it can be operated at room temperature.

As the current InGaAs photodiodes are not sensitive to $2\text{ }\mu\text{m}$ light, ‘extended’ InGaAs photodiodes may be used instead. These diodes are extended in the sense that their cut-off wavelengths are above $2\text{ }\mu\text{m}$. To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium than normal; however, this extra indium comes at the expense of the structural integrity of the lattice structure. This can cause the quality of the photodiode to be degraded. Previous experiments have been limited by the quantum efficiency of the extended InGaAs photodiode that was used [7,8]. To obtain low enough electronic noise, the authors of [7] used the photodiodes without a bias. A photodiode with a quantum efficiency of up to 95% has been reported.

Alternative materials that could be used include HgCdTe and InSb. HgCdTe has potential, however there are still technical challenges to overcome in creating diodes that work at???. The properties of an InSb detector with a high reported QE is reported in chapter ??, however this detector would not be suitable for use in a gravitational wave detector.

In this chapter, the relationship between quantum efficiency, reverse bias and noise in Extended InGaAs Photodiodes is explored. A selection of extended InGaAs photodiodes were tested with the aim of understanding general trends in the behaviour of extended InGaAs diodes. It was found that to get a high quantum efficiency and linearity, the diodes must be operated with a bias. However, if the diodes were used with a high bias then the dark noise of the diode would exceed the shot noise level of the detected light. The dark noise could be reduced significantly by cooling the diodes and so future detectors may feature cooled read-out diodes. Generally speaking, the longer the cut-off wavelength, the worse the diodes noise was. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a diode made from extended InGaAs can be reduced by orders of magnitudes by optimising this. However, at the time of writing (2020), commercially available InGaAs photodiodes for detecting 2 μm light that are suitable for use in a gravitational wave detector do not exist.

1.3 What is an Extended InGaAs Photodiode

A photodiode is a square law detector and so it produces a current that is proportional to the power of the light it is detecting. Usually, they are of the order 1 mm in

diameter. They are two-terminal devices, with an anode and a cathode. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) structure. PIN photodiodes are used in applications where there is milliwatts of light power.

To understand a PIN photodiode, it is useful to first consider a simpler PN photodiode. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the P-doped region, holes (the majority charge carriers) are free to move whereas in the n-doped region electrons (the majority charge carriers) are free to move.

At the interface between the p-doped and n-doped region, majority carriers from each side diffuse across it. Then, the minority charge carrier combines with a majority charge carrier, and so no net current will flow out of the device. When in equilibrium, there are no free carriers in the p and n regions, and so these regions become charged. Due to the diffusion of majority charge carriers across the interface, there is an electric field between the p-doped and n-doped region. The region of the device containing no free carriers is called the depletion region.

When a photon is incident on a pn junction, it will create an electron-hole pair. If this photon is absorbed within the depletion region of the diode, then due to the electric field, the electron and hole will be separated. These electrons/holes will generate a current in the circuit connected to the diode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will not be separated and will recombine. These photons will not contribute to the current in the circuit.

To widen the depletion region, an external voltage can be used. When *reverse bias* is applied to a photodiode – **that is a positive voltage across the cathode and anode** –, there is a larger electric field difference at the interface between the p-doped and n-doped region. This increases the volume of the photodiode results in absorbed photons contributing to the photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p and n-doped region. An intrinsic semiconductor has no charge as it has an equal number of excited electrons and holes. An electric field will be present in a larger volume of material and so more photons will be converted into electrons which leave the photodiode diode.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, so that the InGaAs can be lattice matched to InP substrate on which the photodiode is grown on. At this level of indium doping, the InGaAs photodiode will be suffer from few mechanical defects and so the diodes created with it will be of high quality. Typically they are sensitive up to wavelengths around 1.6 μm .

Longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 μm . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which its grown. Lattice mismatching can arise due to a strained crystal structure, and since the extra indium in an extended InGaAs photodiode causes this [9] it is hard to fabricate a low noise extended InGaAs photodiode [10]. Above

53% indium, a buffer layer of InGaAs with less indium in between the absorbing layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [11]. Lattice mismatching is the source of the large amounts of dark current found in extended InGaAs as well as the $1/f$ noise.

similar circuit model?

1.4 What is Needed from the Read-out Photodiode in a Gravitational Wave detector

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that to achieve the baseline level of ~ 10 dB of squeezed light that future gravitational wave detectors will use, losses in the interferometer have to be extremely low as losses degrade the amount of squeezing achieved.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that they are not limited in the detection band by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [12], so the $1/f$ noise must be below the shot noise of the photo-current above this frequency.

The $1/f$ noise could be avoided by using a heterodyne detection, however the shot noise would increase if the mixing of the modulation and demodulation waveforms was not time independent [13] and this limits the choice of waveforms. These schemes recover increase the SNR by increasing the signal, however the use of squeezing means that this approach would still not be sufficient as the noise would still be enhanced compared to a homodyne readout. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [14].

Current gravitational wave detectors operate in the 1-10 mW range at the output [15] to ensure that the junk light from other modes is a small portion of the total signal. Thus, any new photodiode should be able to produce ~ 10 mA of photocurrent.

The properties were investigated in a selection commercially available photodiodes. The photodiodes that were tested are listed in table ???. According to each of the diodes datasheets, none of them have 99% quantum efficiency; however, the quantum efficiency of a diode can be improved by increasing the bias voltage (see Section ??). Increasing the bias increases the dark current and dark noise of the photodiode and so for each diode a maximum reverse bias voltage was found. The bias voltage also affects the saturation limit of the photodiode. If a diode was found to have a low saturation threshold (~ 1 mA), then the diode is not fit for use in a gravitational wave detector.

1.5 The effect of reverse bias on an Extended In-GaAs Photodiode -FD10D

To demonstrate the effect that reverse bias has on the quantum efficiency of a diode, the Thorlabs FD10D was used. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: One where the reverse bias and the response of the diode were measured at DC, and a lock-in style measurement where the quantity $\frac{\Delta\eta}{\Delta V_B}$ was measured. The quantum efficiency should be independent of frequency if the diode is operating in a linear regime, and so the two styles of measurements should yield the same outcome. To prove that the increase in signal is a result of increasing quantum efficiency and not some other mechanism, for instance an avalanche effect, the noise of the photocurrent was compared to the level expected for shot noise at a bias voltage that yielded an acceptable level of dark noise (see Section ?? for details.).

1.5.1 Apparatus

A 1550 nm wavelength laser (Pure Photonics PPL) was used to illuminate the photodiode (FD10D). The beam was split by a balanced beam splitter so that the light could be amplitude stabilised. The in-loop diode was a high performance InGaAs photodiode (C3061GH), and the out-of-loop diode was the diode under test. The laser was amplitude stabilised by actuating on the laser diode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

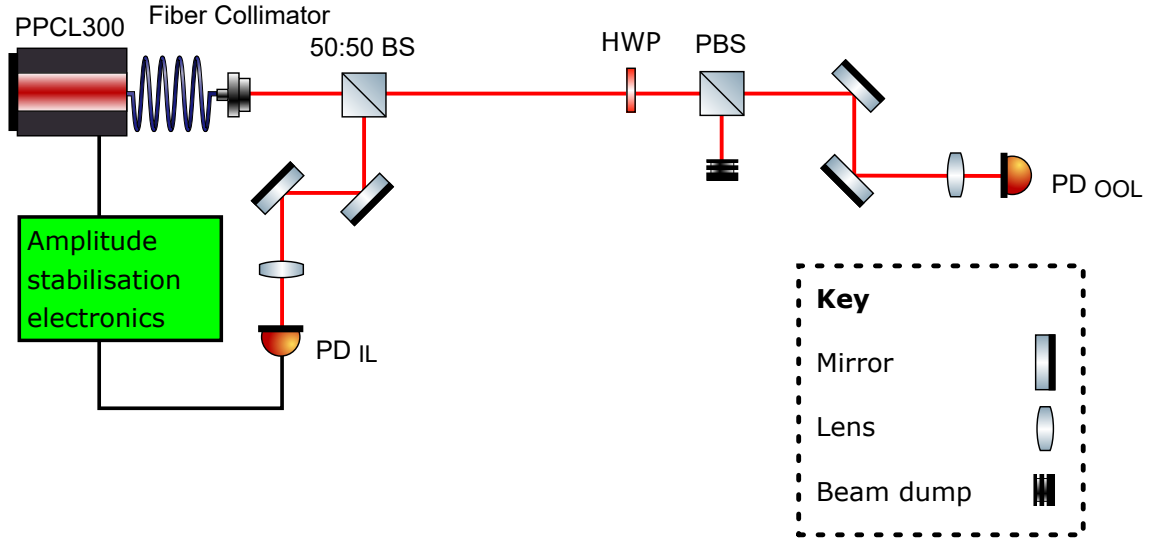


Figure 1.1

1.5.2 Method

The FD10D was exposed to a variety of different beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the beam size and beam power. As the FD10D is 1 mm, the maximum beam width would be around $330\text{ }\mu\text{m}$ in order for less than 1% clipping losses. The powers used were in the milliwatt range as these types of beam powers are the ones that a gravitational wave detector would operated with. The methods in this section can only be used to measure relative increases in quantum efficiency, measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. By measuring the increase in photocurrent at DC it is straight forward to determined the increase in quantum efficiency. Alternatively, one can perform a lock-in style measurement. If the quan-

tum efficiency depends on the reverse bias of the photodiode, then modulating the reverse bias, V_B at a frequency f will result in η modulating at f , and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode at a reverse bias V_B and θ is the phase between the input and output modulation.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see section ??), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. This was done to ensure that the increase in photocurrent was due to a real increase in quantum efficiency and not due to another mechanism e.g. an avalanche process. The expected level of noise, n , of the photocurrent, in units of $\text{A}_{\text{rms}}/\sqrt{\text{Hz}}$, is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (1.1)$$

where q is the charge of the electron, $I_{\text{OOL/IL}}$ is the current through the out-of-loop and in-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop diodes and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser}^2 is the noise of the analyser. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the linearity of the diode could be investigated.

At this bias, the quantum efficiency was calibrated using two other power meters. The power incident onto the photodiode was measured and the power reflected by the window at small angles was measured.

1.5.3 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 1.2. For all beams intensities that the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V to 1.8 V, the stated maximum reverse bias in the datasheet of the FD10D. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced a greater increase in η than beams with a higher intensity, however the quantum efficiency may not be uniform over the diode – near the edges η is greater. The difference between the DC and AC response could only be described by a non-linear process. These effects are small, and Figure 1.2 gives a clear incentive to increase the bias voltage.

Figure 1.3 shows the photocurrent measured has the noise level predicted by Equation 1.1, meaning that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. This measurement was performed with 4 different photocurrents, ranging from 7.5 mA to 15.5 mA, and over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 1.4. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak applied at 1111 Hz and the level of shot noise.

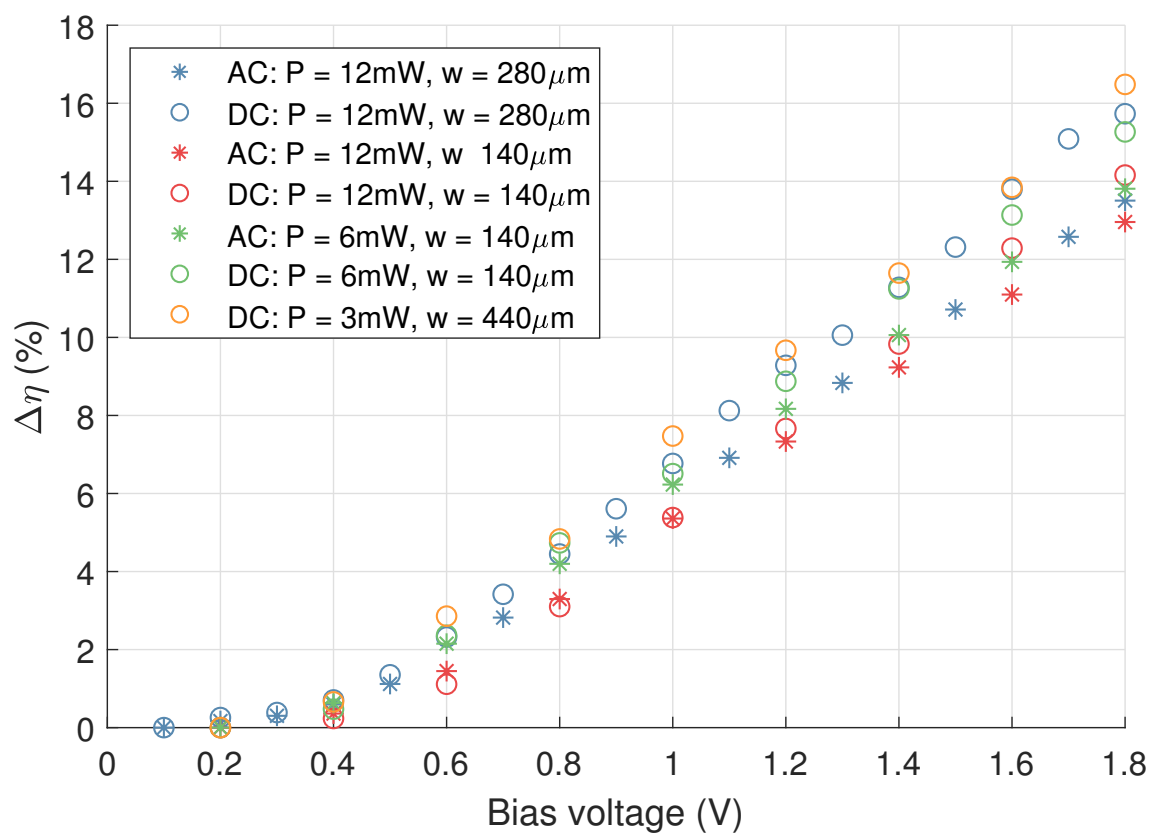


Figure 1.2

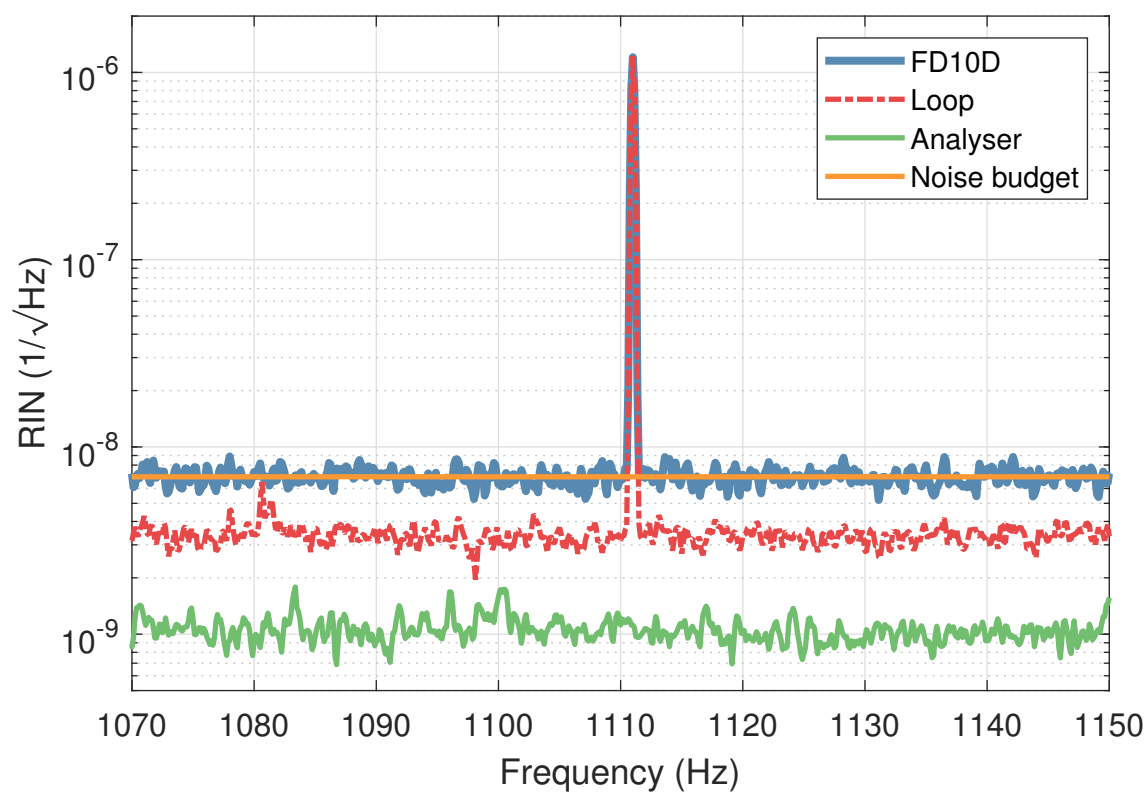


Figure 1.3

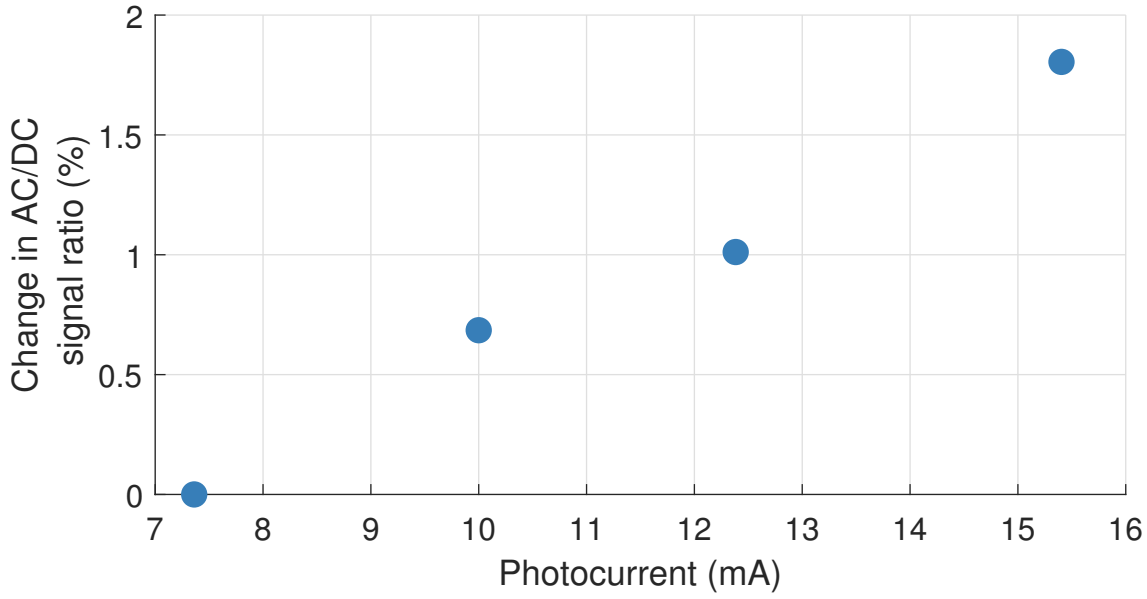


Figure 1.4

1.5.4 Saturation Limit for some laser components photodiodes

The laser components photodiodes were exposed to a light signal of $RIN = 12e-6$. Their bias was set to the maximum that was compatible with $1/f$ noise being below shot noise (see Section ??). The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 1.5. A decrease in AC signal indicates that the diode is saturating. These diodes cannot handle the power levels required by for use in an GW interferometer. This problem was not evident in the FD10D (see figure 1.4).

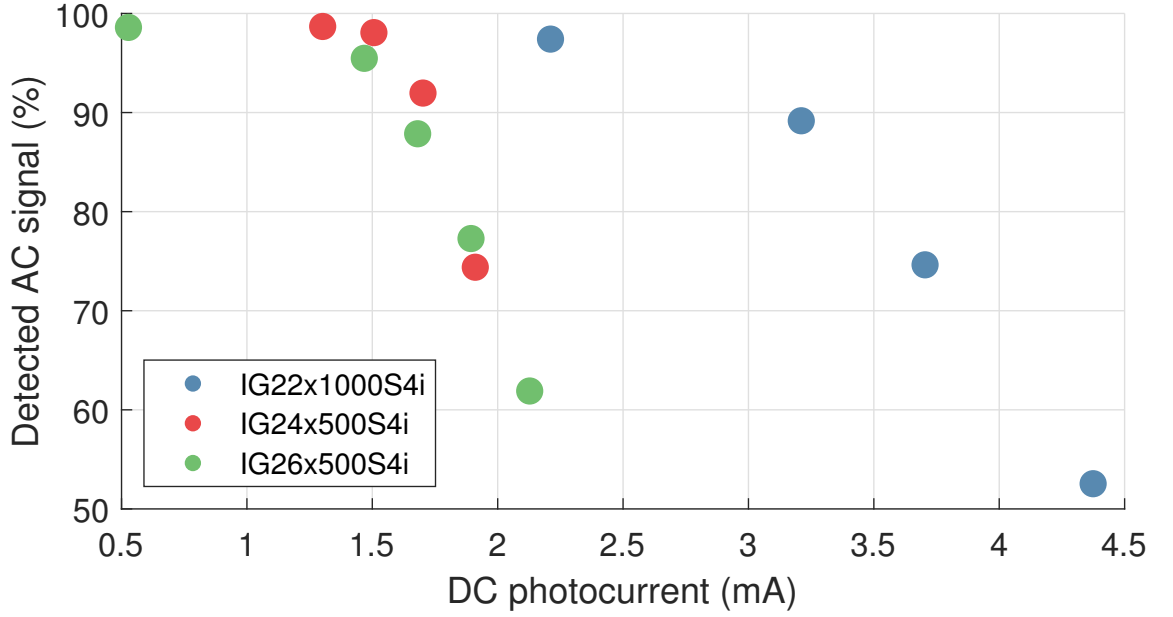


Figure 1.5

1.6 Dark Current and Dark Noise in Extended InGaAs Photodiodes

The dark current and dark noise of extended InGaAs photodiodes has been a limiting factor in previous quantum optics experiments [7, 8]. The dark current of these photodiodes depends on their reverse bias. The larger the reverse bias, the higher the dark current. The dark current of extended InGaAs photodiodes is high because of the relatively large amounts of dislocations due to lattice mismatch between the InGaAs and the substrate that it is grown on.

There is a balance to strike between the quantum efficiency and dark noise. It is desirable to have as large a reverse bias as possible as this increases the quantum efficiency and saturation limit of the photodiode (see section ?? or figure ??). The

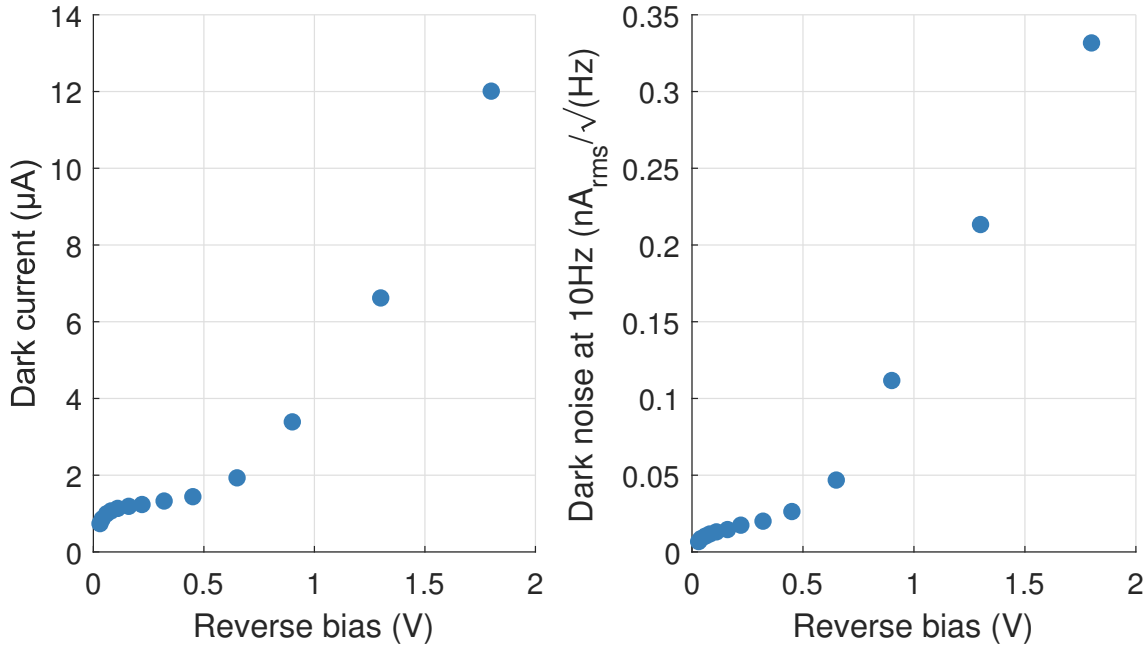


Figure 1.6: FD10D dark current and dark noise

dark noise of the photodiode should be below the shot noise level of the light that will be detected. As a bench mark, the dark noise of the photodiodes at **10 Hz** is compared to the dark noise of 10 mA of photocurrent.

1.6.1 Noise in photodiodes

The noise of the current made by the photodiode can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R} + \left(\frac{i_t}{f^\alpha}\right)^2} \quad (1.2)$$

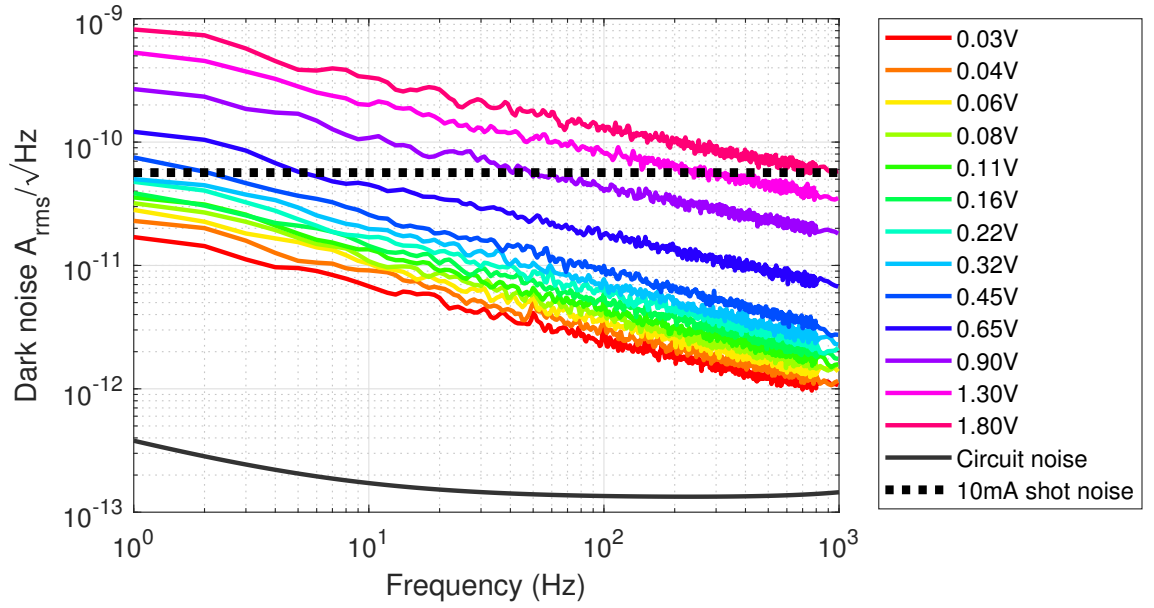


Figure 1.7: FD10D dark noise

1.6.2 Room Temperature

1.6.2.1 FD10D dark current and Dark noise

1.6.2.2 Comparisson of all the InGaAs Photodiodes

To compare different photodiodes, consult figure 1.8.

1.6.2.3 Variability between ‘identical’ Hammamatsu Diodes

1.6.3 Cooling Experiment

The typical low temperature damage threshold for extended InGaAs photodiodes is -40°C . To provide controlled cooling to the photodiode, the photodiode was glued (using Cryo-Varnish) to a block of aluminium, shown in figure 1.10. This block has a copper rod with a diameter of a few mm soldered to it, and this rod was inserted into a liquid nitrogen bath. The block of aluminium had power resistors to provide

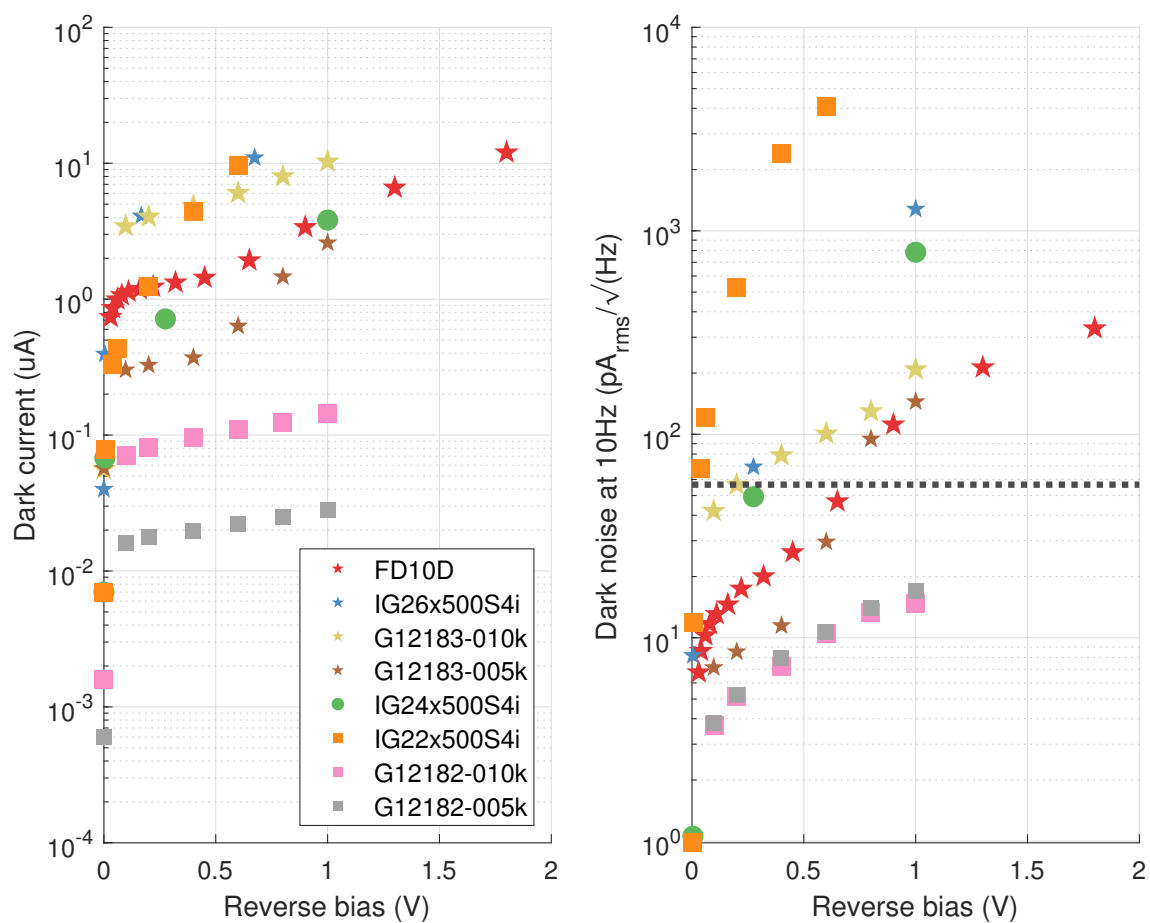


Figure 1.8

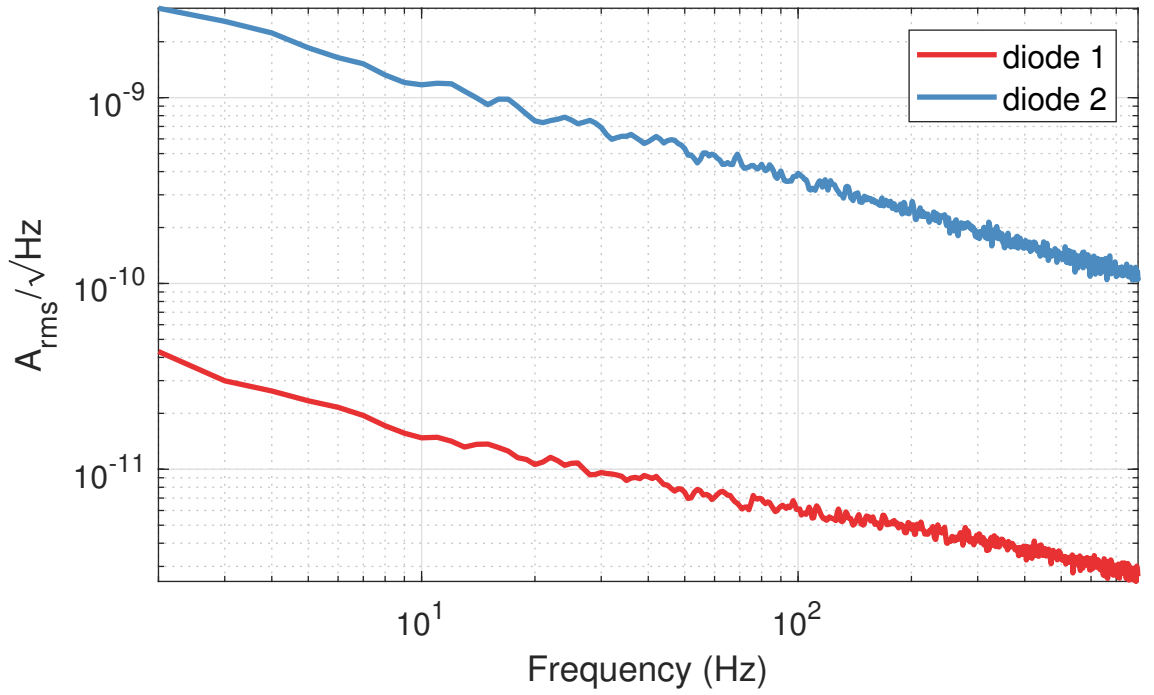


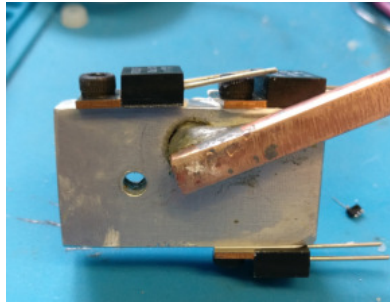
Figure 1.9

heating so that temperature of the diode could be controlled. The block is shown in figure 1.10. A PID loop was implemented using a pt1000 temperature sensor and a multimeter (keithley 2000) configured to make a 4 wire resistance measurement. A labview program read this signal and performed calculated the error signal to be sent to a programmable power supply (Temna 72-2540) connected to the power resistors.

See appendix ?? for dark noise as a function of bias. The dark current as function of reverse bias voltage is shown in Figure 1.11.

1.6.4 Forward Bias Characteristics of IG24

The ideality factor of a photodiode will



(a)



(b)

Figure 1.10

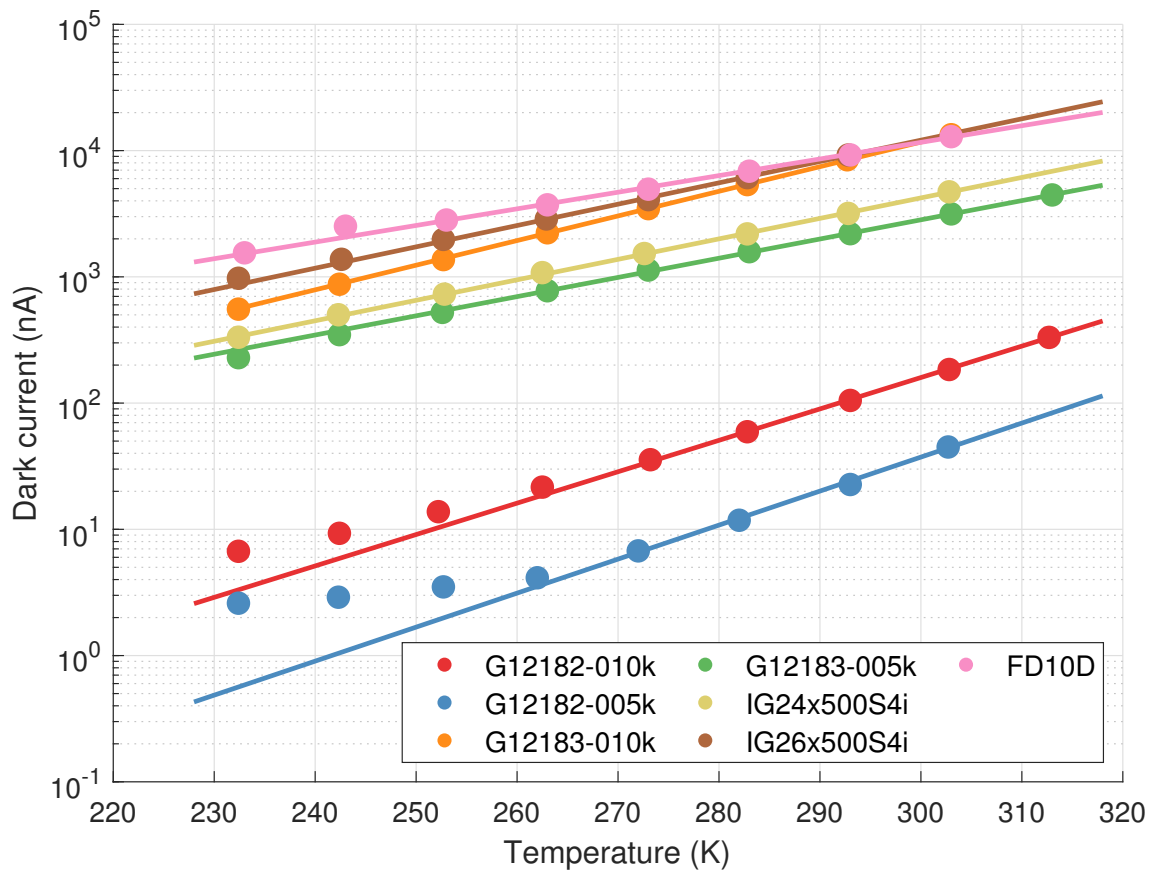


Figure 1.11

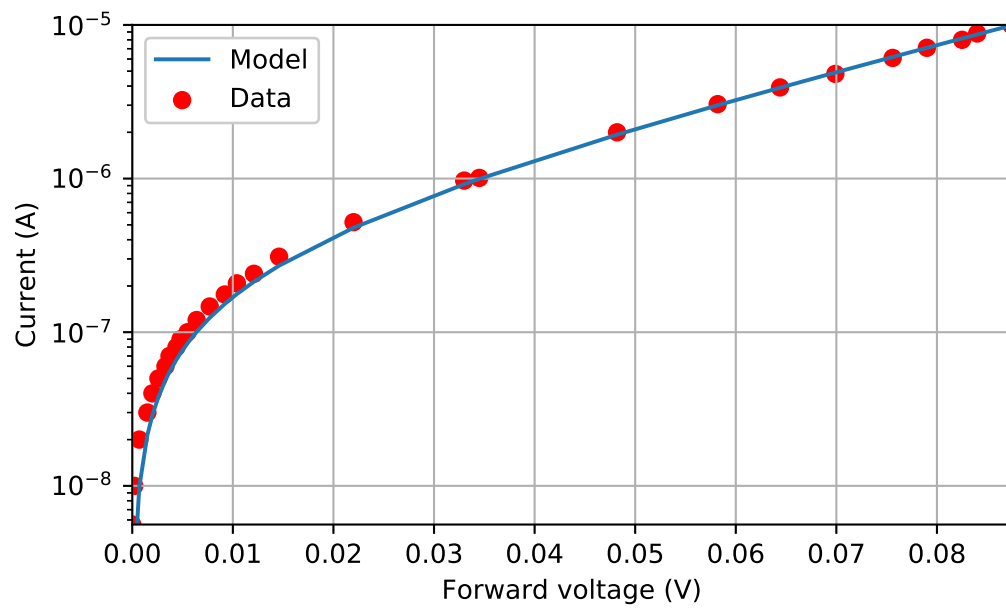


Figure 1.12: $I_0 = (360 \pm 10) \text{ nA}$, $n = 1.02 \pm 0.01$

1.7 InSb testing

1.8 Discussion

1.9 Conclusion

Appendix A

Extended InGaAs Diodes Noise Spectrums

A.1 Bias Section

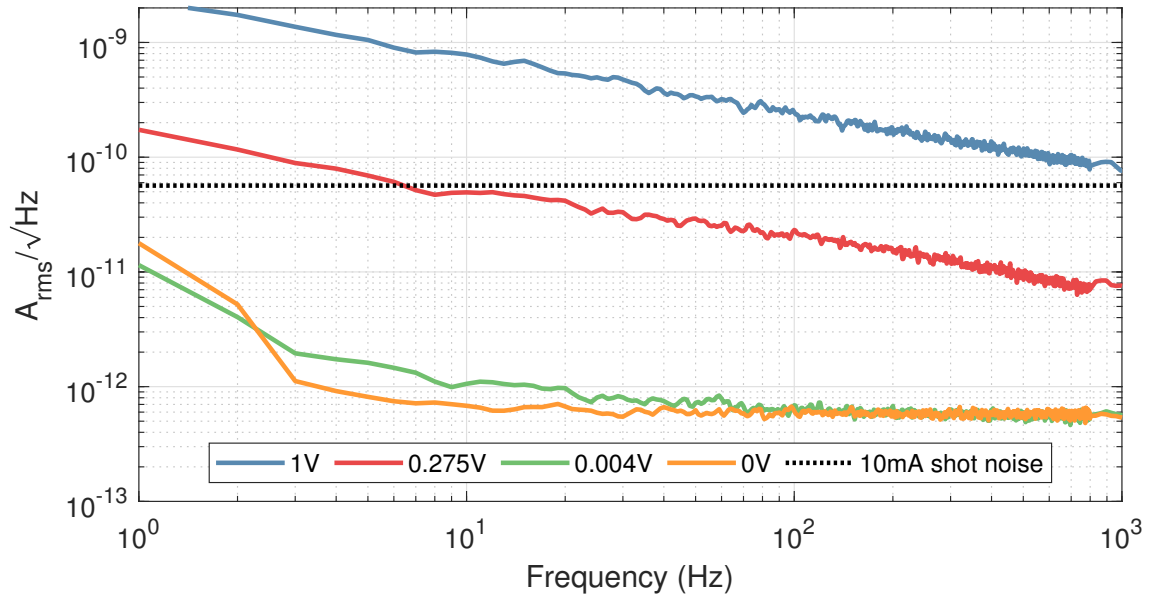


Figure A.1: IG24

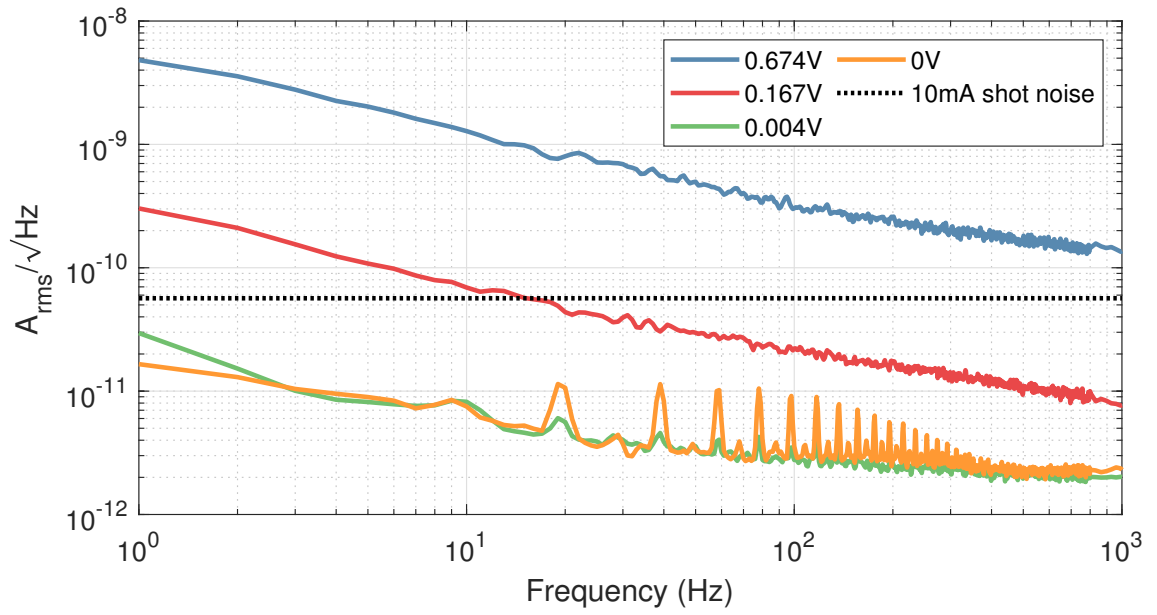


Figure A.2: IG26

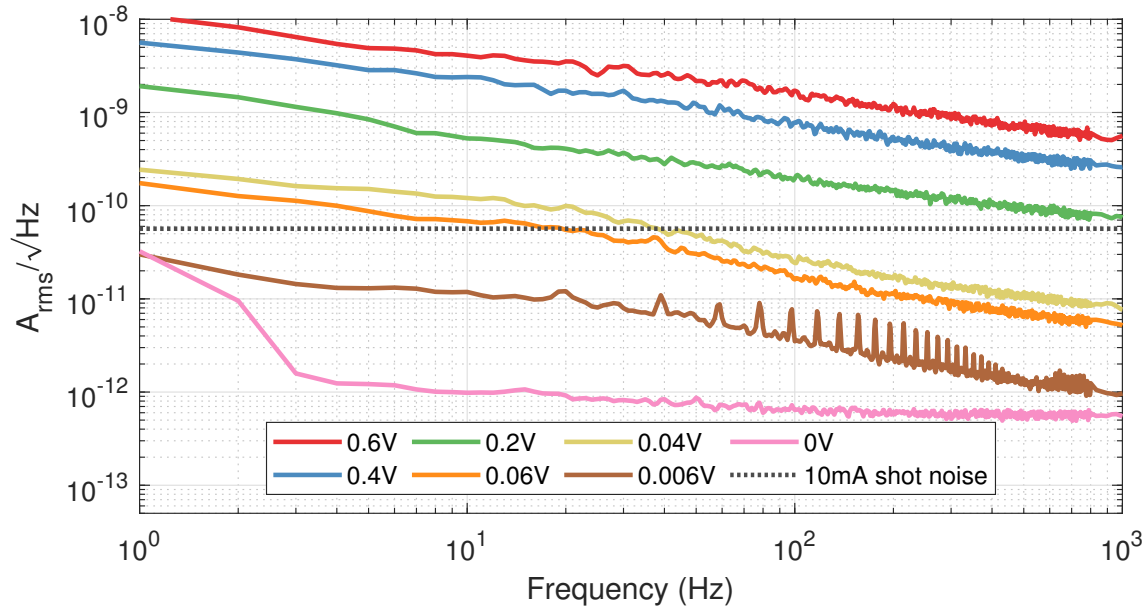


Figure A.3: IG22

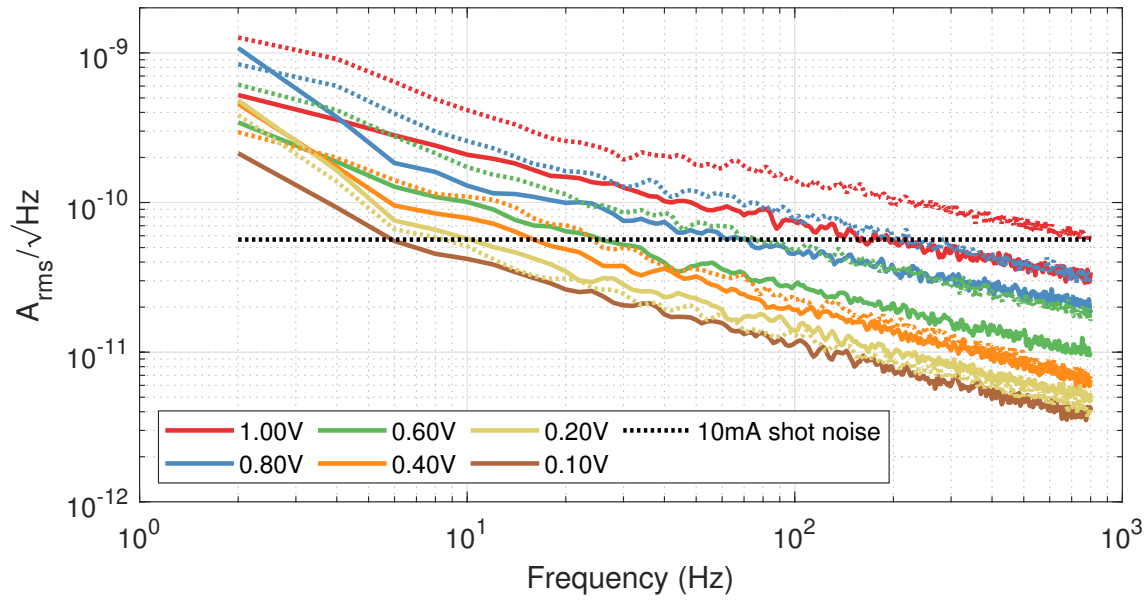


Figure A.4: G12183-010k

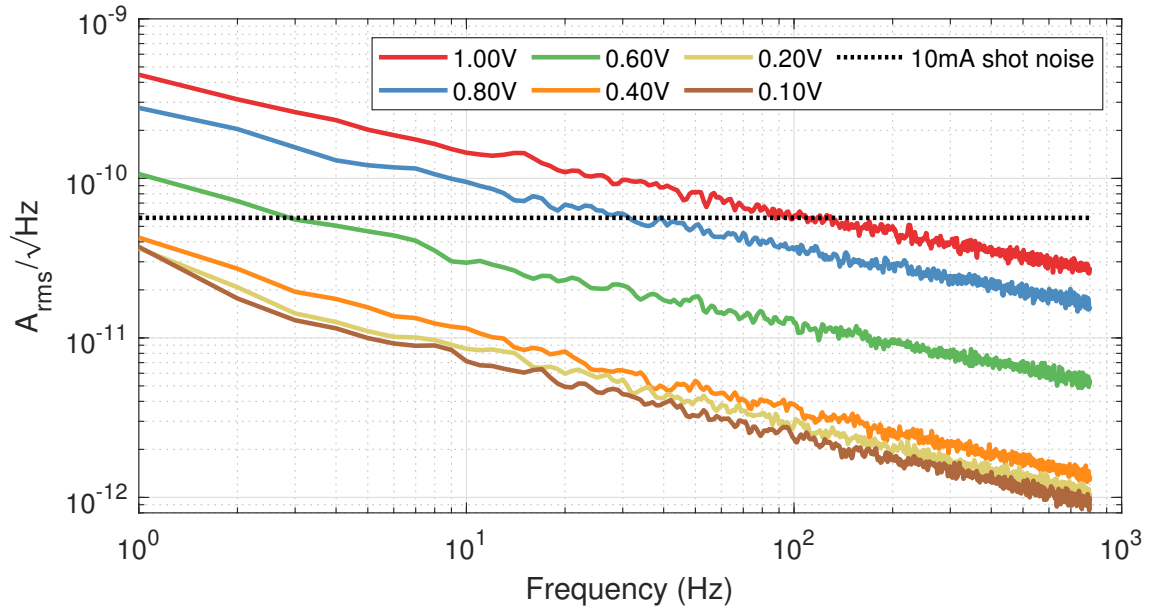


Figure A.5: G12183-005k

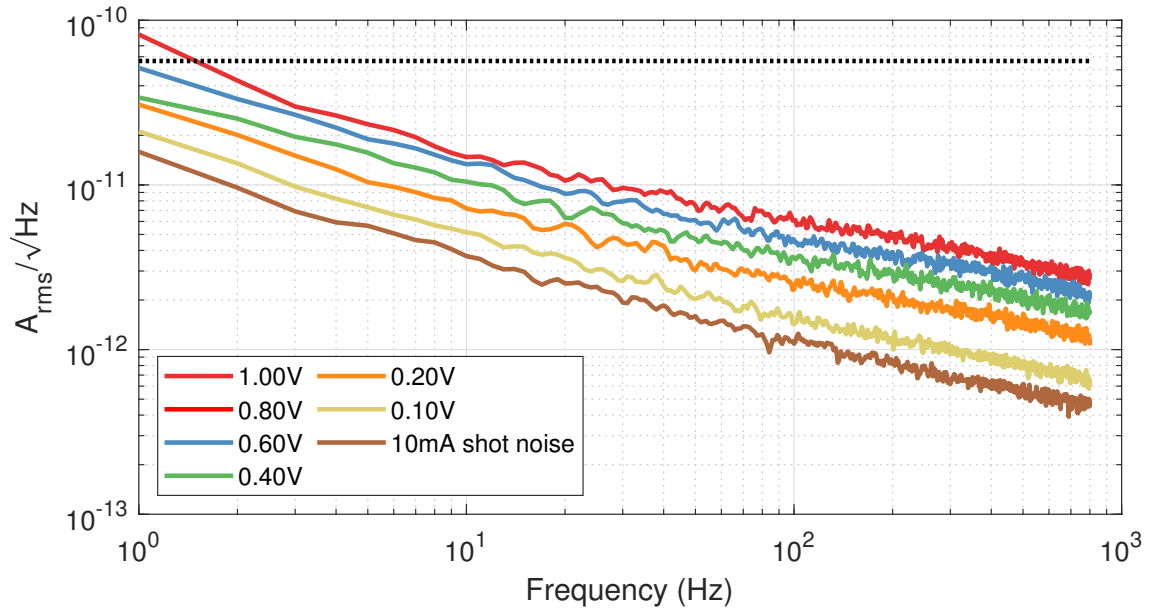


Figure A.6: G12182-010k

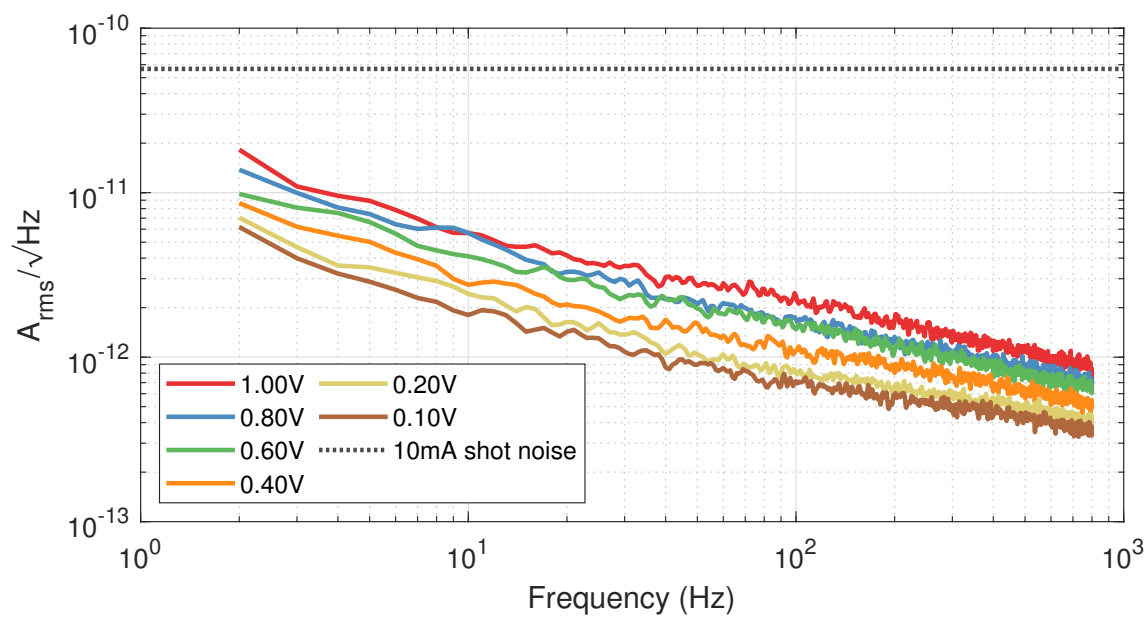


Figure A.7: G12182-005k

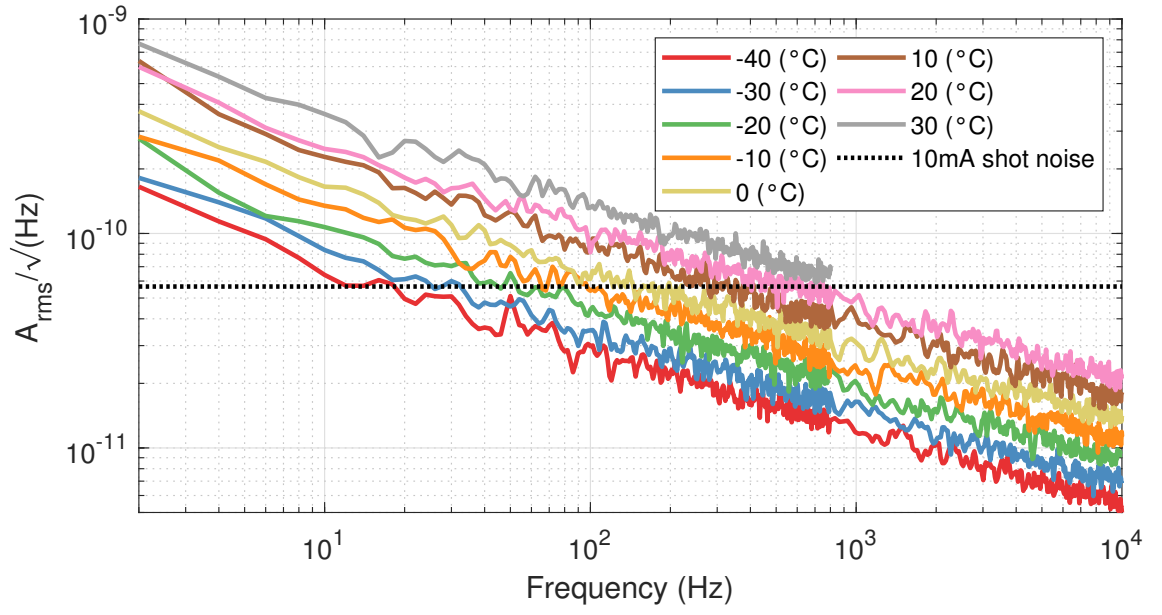


Figure A.8: FD10D 1.8V reverse bias

A.2 Temperature Section

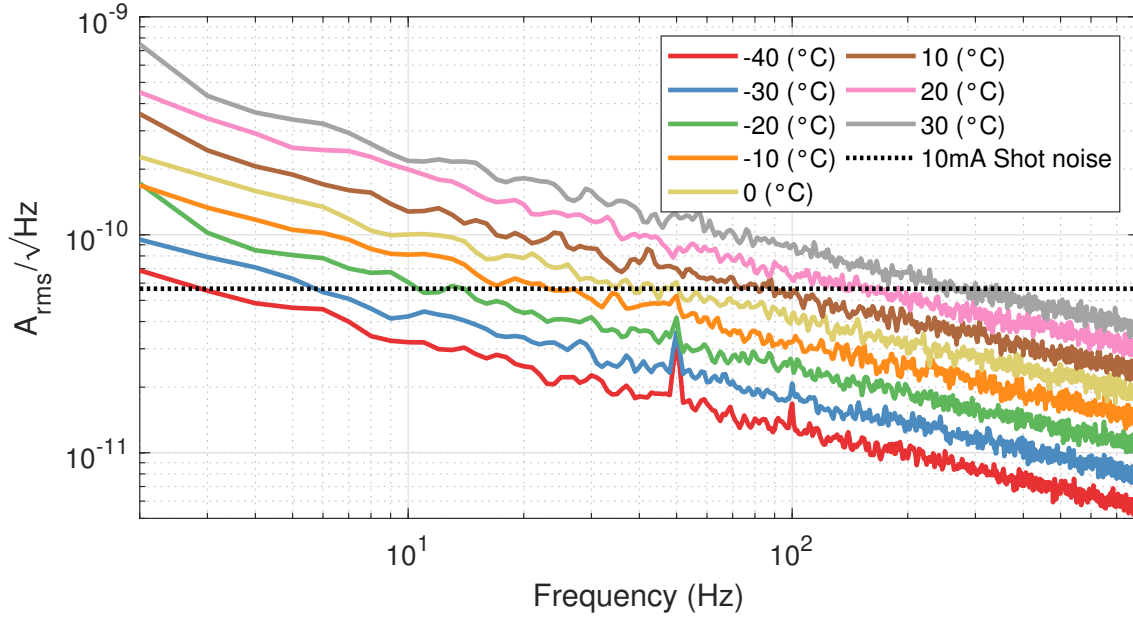


Figure A.9: G12183-010k 1.0V reverse bias

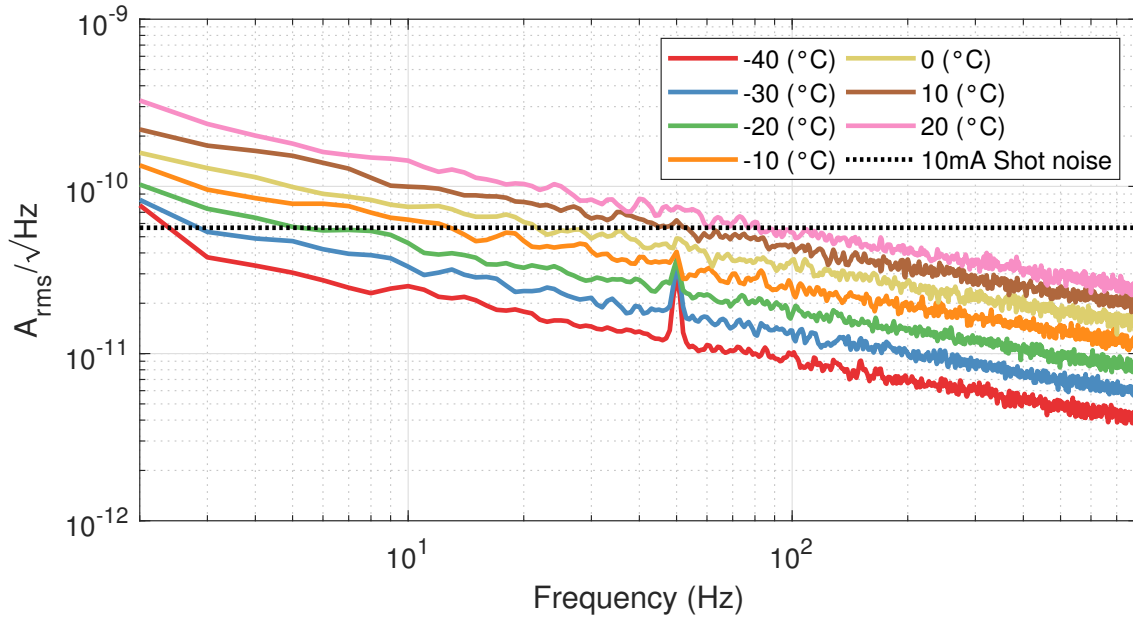


Figure A.10: G12183-005k 1.0V reverse bias

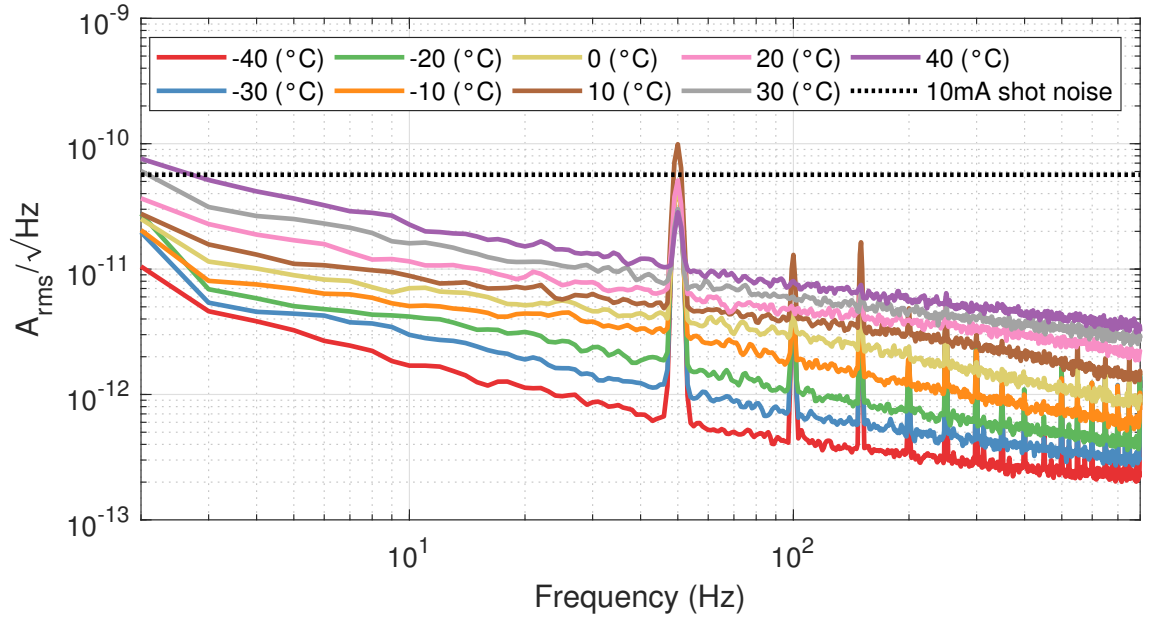


Figure A.11: G12182-010k 1.0V reverse bias

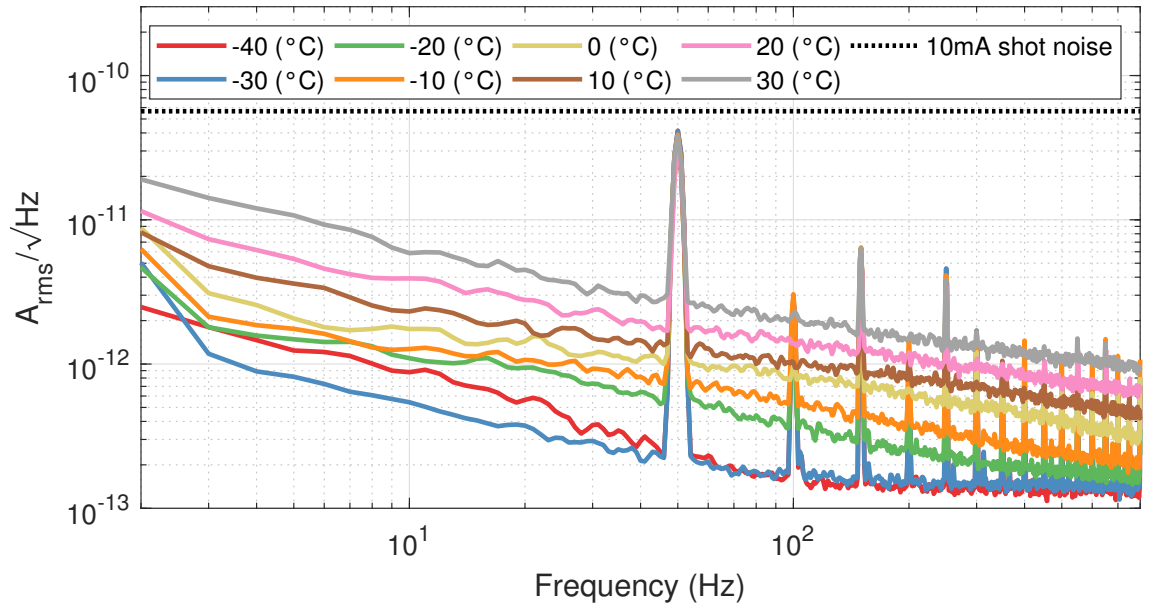


Figure A.12: G12182-005k 1.0V reverse bias

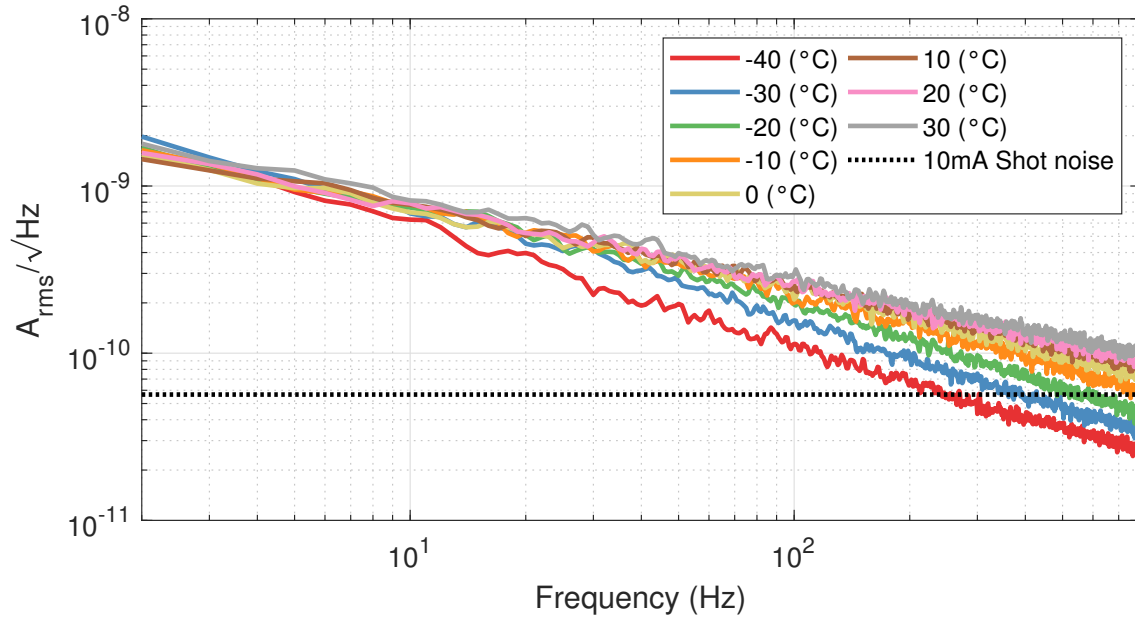


Figure A.13: IG24x500S4i 1.0V reverse bias

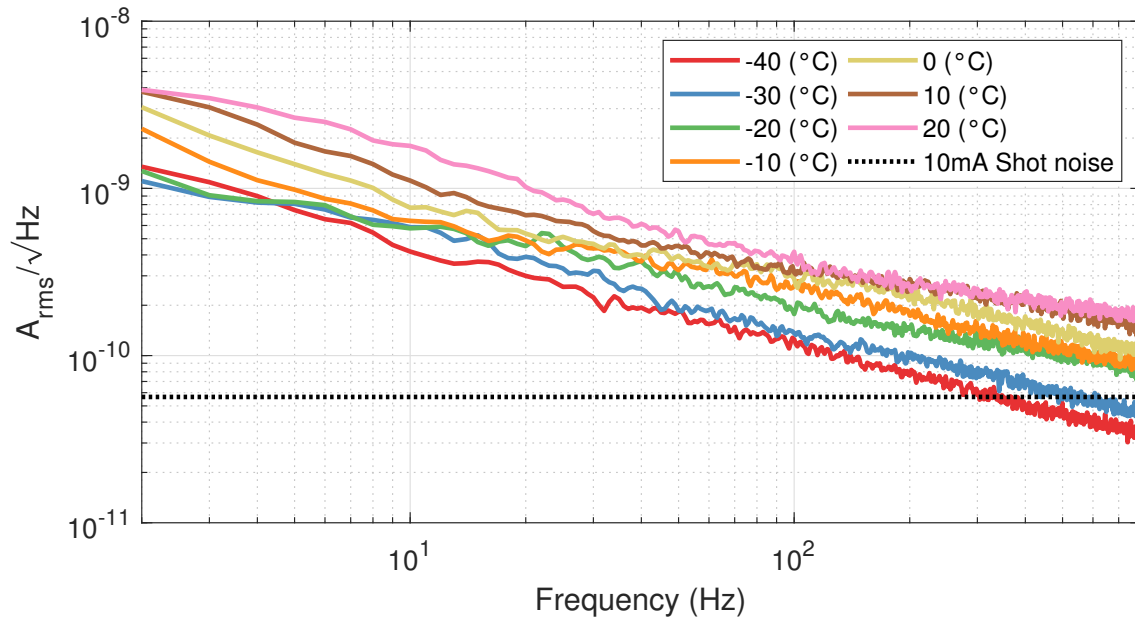


Figure A.14: IG26x500S4i 1.0V reverse bias

Appendix B

Photodiode Circuits

B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes

A circuit which allowed for the reverse bias to be set and for the easy exchange of photodiodes was constructed. To measure small amounts of current noise, a transimpedance amplifier with a large gain is needed. As the dark current for these diodes is of the order of 1mA, the gain of the transimpedance amplifier was chosen to be 1×10^6 so that signals of the order 1 V would be produced, and so a $1 \text{ M}\Omega$ resistor was used as in the feedback path.

Operation amplifiers have current noise at their inputs, voltage noise at the their outputs and the feedback resistor will generate Johnson-Nyquist noise (e.g. [16, Figure 8.58]). Designers of OP amps, generally, improve the OP amps output voltage noise at the expense of the input current noise or vice versa and so there is not one-size-fits-all op amp. In this application, as the current noise that is to be measured

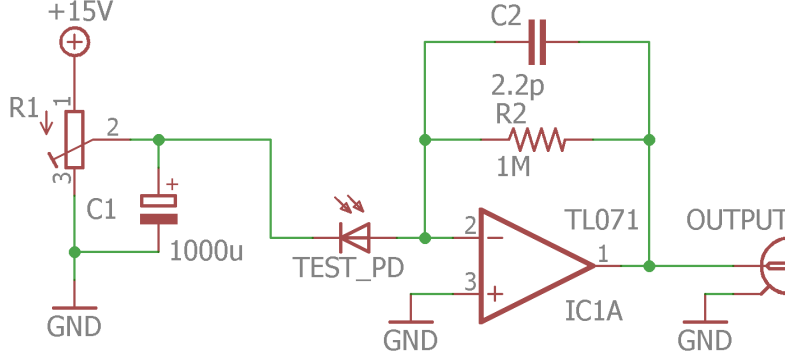


Figure B.1: Circuit used to measure the dark noise of photodiodes, represented by TEST_PD, under different reverse biases. The transimpedance amplifier has a $1\text{M}\Omega$ (R2) in parallel with a 2.2pF capacitor (C2) in the feedback path of the TL071 op amp (TL071P). The reverse bias was generated by a potentiometer (R1) connected to an LM7815 voltage regulator. This voltage was low passed using C1

is small and the feedback resistor is large, a TL071, a JFET style op amp with low input current noise, was selected. As the photodiode has capacitance and the op amp has finite bandwidth, a 2.2pF capacitor in parallel to keep the circuit stable.

Several methods of providing the reverse bias were tested. The first of these derived the bias from a LM7815 regulator in combination with a variable resistor configured a potential divider; this circuit is shown in Figure B.1. A similar circuit which was used in later experiments is shown in Figure B.2.

To verify that this was a sufficiently quiet bias voltage source that does not contaminate the measurements with noise, an ultra-quiet bias reference was used to cross check a measurement; this circuit is shown in Figure B.2. By filtering the voltage made a reference voltage IC (AD587JNZ), this circuit provided a bias with

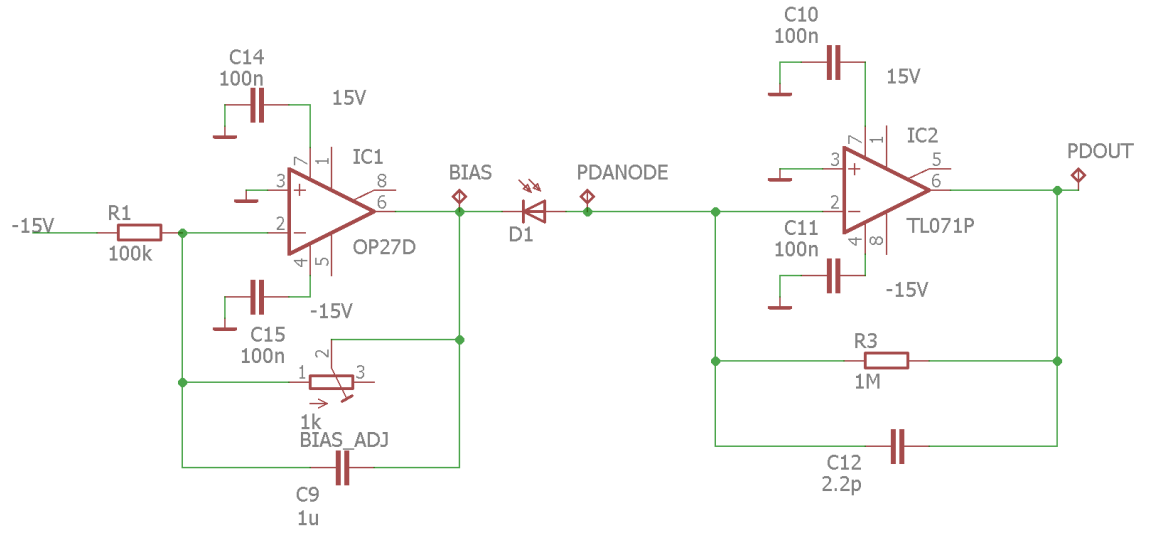


Figure B.2: Circuit used to measure the dark noise of photodiodes. The transimpedance amplifier used is identical to the one shown in Figure B.1. The bias voltage was generated using an OP27 op amp (OP27D) configured with so that the gain between the inverting input and the output of OP27D was > 1 .

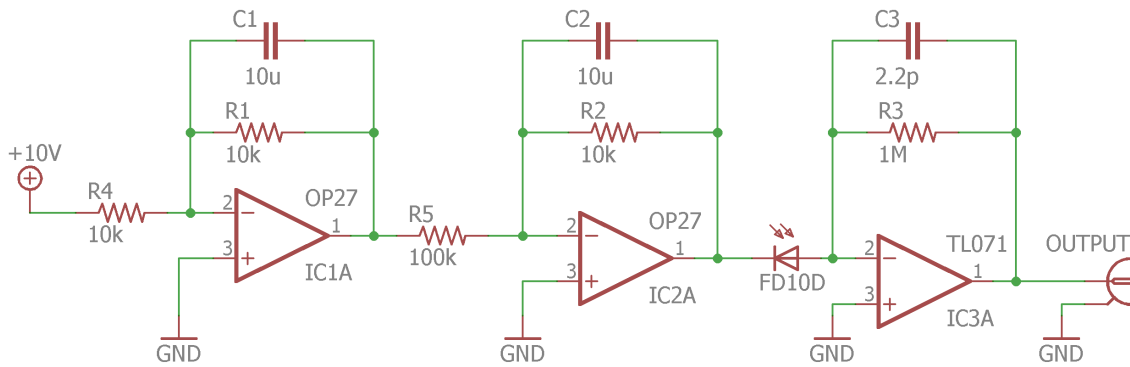


Figure B.3: Circuit used to check that bias voltage noise was not contaminating dark noise measurements. To reach $5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz, a 10V low noise reference IC (AD587JNZ) was used to generate the reverse bias. The output of this IC was filtered by two low pass filters (IC1A and IC2A). The transimpedance amplifier used is identical to the one shown in Figure B.1

$5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz. The bias of the FD10D (see Section ??) was set to 1 V in both circuits. Figure B.4 shows that the regulator-potentiometer bias voltage source is sufficiently quiet.

B.1.1 Transfer function

The transfer function of the transimpedance amplifiers shown in Figure ?? was measured under a variety of different physical constructions. This circuit's behaviour can be affected by stray capacitance. When constructed from strip board and with the photodiode anode and cathode cables very near each other, there was of the order 10 pF of stray capacitance i.e. it was as if the circuit had an extra 10 pF capacitor between the inverting input of the op amp and ground. By making this circuit on a prototyping board with less stray capacitance between the strips and

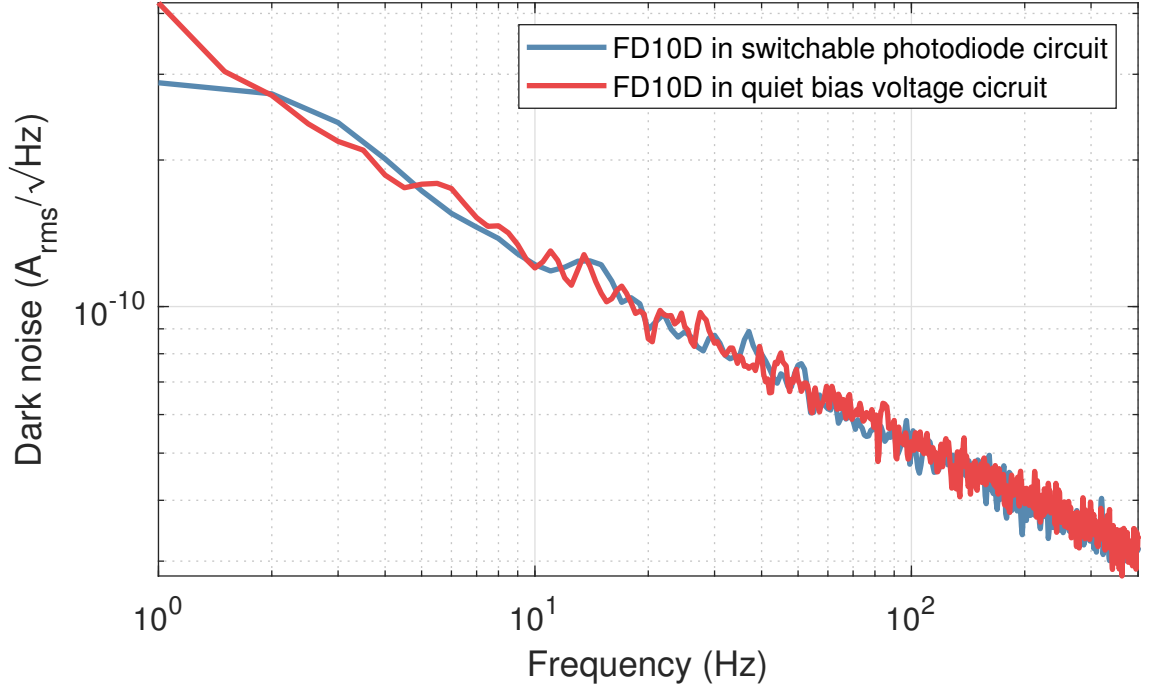


Figure B.4: The FD10D was biased with 1 V in the circuits shown in Figure B.1 and Figure B.2. The low noise fixed bias circuit and the circuit with a settable bias gave the same results, so we can be confident that the circuits shown in Figures B.1 and B.3 will produce measurements that are not contaminated by bias noise.

by keeping the photodiode anode and cathode wires from forming a capacitor, the stray capacitance was reduced. The measurement is in agreement with the LISO simulation of this circuit. This was cross checked using the method described in Section B.4.

B.1.2 Noise measurement

The dark noise of the circuits described in Section B.1 is shown in Figure B.6. The shot noise level of a typical photocurrent in a gravitational wave detector is several hundred times larger than the noise of this circuit. To measure noise of the transimpedance amplifier, instead of a photodiode, a 470 pF capacitor to was connected between the inverting input of the op amp and ground.

B.2 Transimpedance Amplifier for Making Shot Noise limited Measurements of 10 mA Photocurrents

A photodiode circuit which is shot noise limited over a up to several 100 kHz is shown in Figure B.7. A DC bias was created by low-passed bench-top lab power supply. An AC component could be added onto the bias via a summing OP-amp. The FD10D was connected to a transimpedance amplifier that whose electronic noise was below the shot noise level for ~ 10 mA currents. The noise of this circuit is shown in ... the transfer function of this circuit is shown in...

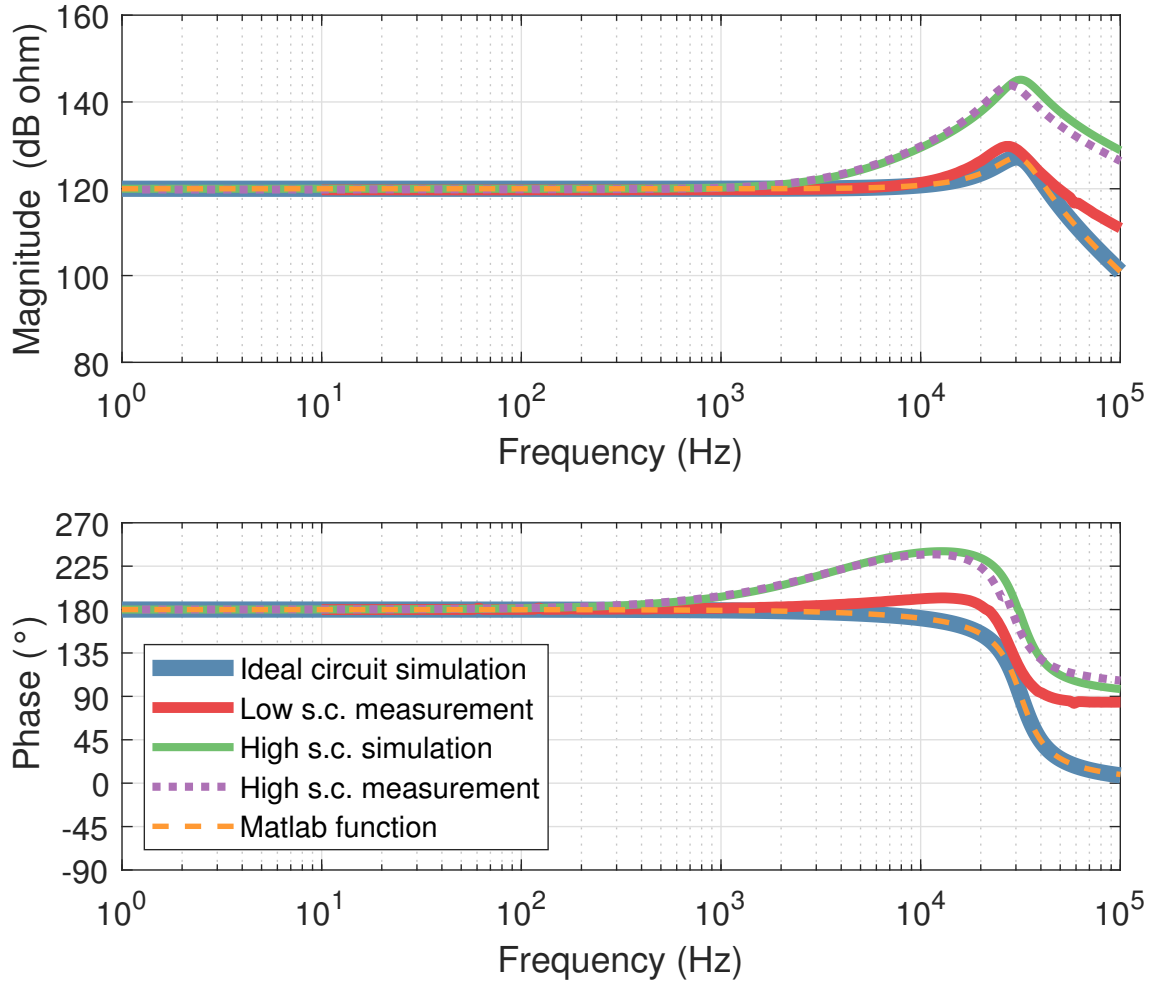


Figure B.5: The transfer function between a current into the inverting input of the transimpedance amplifiers designs shown in Figures B.1B.3 and B.2 was measured and simulated using LISO. A MATLAB function (see Section B.4) was used to check the measurement and the simulation. This circuit is sensitive to stray capacitance (s.c.) at the inverting input of the TL071. The stray capacitance was reduced by changing the layout of the circuit.

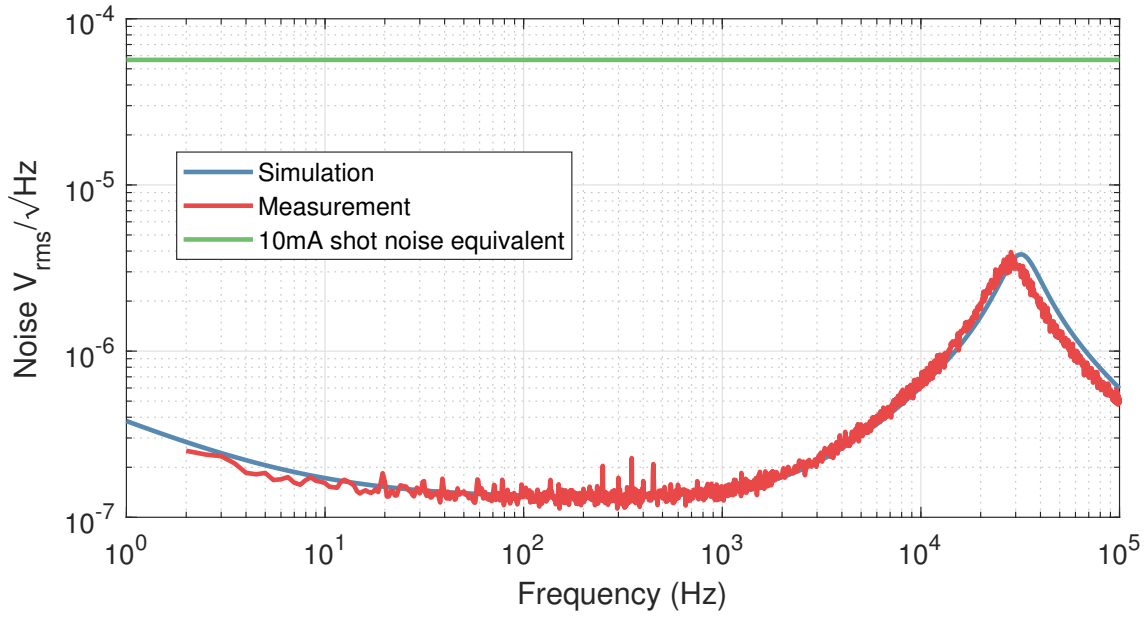


Figure B.6: At low frequencies, the noise of the circuit design shown in Figures B.1B.3 and B.2 is dominated by the low frequency voltage noise of the TL071. Above 10 Hz, the noise of the circuit is dominated by the johnson-Nyquist noise of the feedback resistor. At 30 kHz, the impedance of the feedback network causes the TL071's voltage noise to rise above the resistors noise.

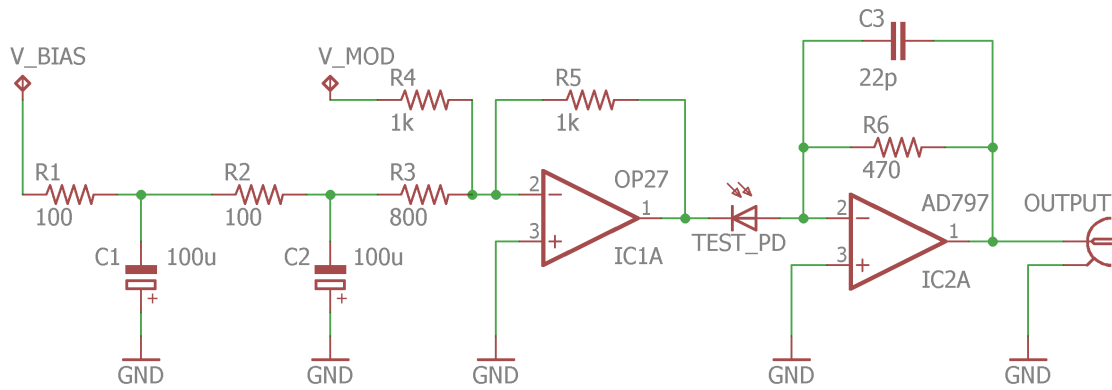


Figure B.7

B.3 Simulations with LISO

The behaviour of circuits was simulated using LISO. This software performs a nodal analysis to calculate the voltages at each point within a circuit. LISO can calculate the noise and transfer function of circuits that include op amps. LISO has a wide variety of op amps in its op amp library and this can be easily expanded by specifying the noise properties, open loop gain and gain-bandwidth frequency.

B.4 Transfer Function Calculation With MATLAB

The open loop transfer function of an op amp is

$$A(s) = \frac{A_0}{\frac{A_0}{2\pi f_{\text{GWB}}}s + 1}. \quad (\text{B.1})$$

The transfer function between the input current to the inverting input and the voltage at the output pin for an op amp made into a transimpedance amplifier is

$$H(s) = Z_{\text{FB}} \frac{A(s)}{1 + A(s)B(s)}, \quad (\text{B.2})$$

where $B(s)$ is the transfer function of the components between the output pin of the op amp and the inverting input of the op amp and Z_{FB} is the impedance seen by the current flowing from the output pin to the inverting input. Without a capacitor in parallel with the feedback resistor $B(s) = \frac{1}{R_{\text{FB}}C_{\text{PD}}s+1}$. This can be problematic as the transfer function of the circuit may have a phase of 180° due to the photodiode's capacitance and the gain bandwidth product of the op amp as it crosses the unity gain point which will cause the circuit to be unstable. To fix this problem, a capacitor

should be added in parallel to the feedback resistor, and so

$$B(s) = \frac{R_{\text{FB}}C_{\text{FB}}s + 1}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.3})$$

The impedance seen by the current is

$$Z_{\text{FB}} = \frac{R_{\text{FB}}}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.4})$$

$H(s)$ can be calculated with the following MATLAB code:

```
function [mag,phase] = photodiodeTransimpedance(cf, rf, cpd, gbw, A, w)
% cf is the capacitor in the feedback loop used to keep the circuit
% stable, and its units are Farads. rf is the feedback resistor
% measured in Ohms. cpd is the photodiodes capacitance (Farads).
    gbw is the
% gain bandwidth of the OP amp, and is measured in Hz. A is the
    Open loop
% gain of the OP amp. w is the angular frequency range that
% the phase and magnitude will be calculated over.
A_s = tf(A, [A/(2*pi*gbw), 1]);
B_s = tf([cf*rf, 1], [rf*(cf+cpd), 1]);
closedLoopAnalytic = -tf(rf, [rf*(cf+cpd), 1])*A_s/(1+A_s*B_s);
[mag, phase] = bode(closedLoopAnalytic, w);
mag = squeeze(mag);
phase = squeeze(phase);
end
```

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